



**Politecnico
di Torino**

N-MON

TEST READINESS REVIEW

Presented by Group - G

Braia Lorenzo Eustachio - s346316@studenti.polito.it

Cavallaro Lorenzo - s346742@studenti.polito.it

Peradotto Simone - s343420@studenti.polito.it



OVERVIEW

- Objectives
- Requirements Recap
- Architecture Recap
- Test Plan
- WBS
- Gantt Chart

OBJECTIVES

Main Goal

- To develop a compact and cost-effective monitoring system that detects and classifies GNSS jamming interference in real-time to protect high-value assets during transit.
- To provide an autonomous system that enhances the security of fleet management by immediately alerting stakeholders of jamming events, allowing for rapid response.

Specific Objectives

- Develop a portable monitoring station deployable on vehicles to continuously analyze GNSS L1 band.
- Integrate advanced signal processing and Artificial Intelligence Algorithms to accurately identify and classify different types of jamming signals
- Create a user-friendly interface to provide clear, immediate visualization of the GNSS signal status
- Ensure an automatic alert to a central control station the moment jamming is detected

USER REQUIREMENTS

UREQ_1	The user shall be able to deploy and operate the system simply by connecting power and network, without complex configuration procedures
UREQ_2	The Control Center shall have a simple, intuitive way to see if GNSS signals are clean or compromised, and what type of interference is disrupting them
UREQ_3	Automatic warning shall be delivered to the user when jamming is detected

FUNCTIONAL REQUIREMENTS

The system shall be able to perform the following operations:

FREQ_1	continuously monitor GNSS signals (L band) multi-constellation
FREQ_2	acquire and then process GNSS signals to extract time-frequency representations
FREQ_3	integrate Artificial Intelligence to correctly identify and classify jamming interference across multiple interference types with different power of jamming signals JSR
FREQ_4	provide the user a simple UI to visualize the output of the AI algorithm and signal status
FREQ_5	automatically send an immediate warning if jamming is present to both the UI and a control center

TECHNICAL REQUIREMENTS

The system will need the following technologies and tools to be implemented :

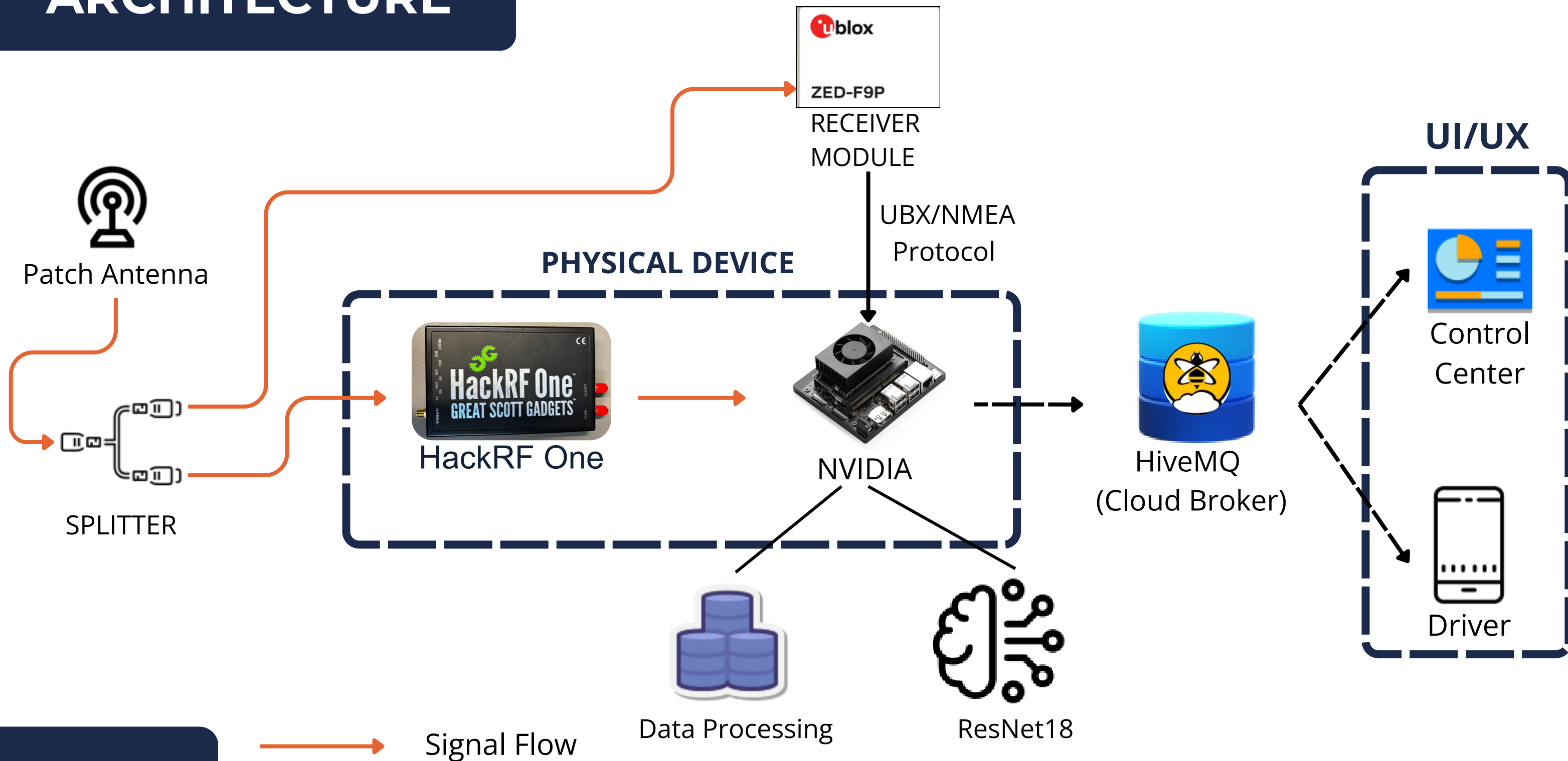
TREQ_1	Multi-band GNSS patch antenna suitable for vehicle-mounted applications, providing reliable signal reception in variable conditions
TREQ_2	An SDR-based RF front-end capable of capturing GNSS signals, with an ADC that provides sufficient sampling rate and resolution for high-quality GNSS signal processing
TREQ_3	computational unit supporting real time AI inference and signal processing operations
TREQ_4	signal processing pipeline to elaborate input data using STFT

TECHNICAL REQUIREMENTS

TREQ_5	test environment with capability to generate synthetic jamming signals with different power of jamming signals JSR
TREQ_6	communication protocols to reliably transmit alerts from the device to remote control station
TREQ_7	web based UI providing real-time status updates
TREQ_8	target cost lower than current laboratory setup
TREQ_9	reduced dimensions compared to existing laboratory setup

SysReq document [here](#)

ARCHITECTURE



TEST PLAN

The test plan is organized into 4 test areas, each derived directly from our requirements. Every test has a unique ID, maps to specific requirements, defines the exact procedure, and states a quantifiable pass criterion

AREA 01

Hardware
&
Signal Acquisition Tests

UREQ_1, FREQ_1,
FREQ_2, TREQ_1,
TREQ_2

AREA 02

AI Model
&
Classification Tests

FREQ_3, TREQ_3,
TREQ_4, TREQ_5

AREA 03

Communication
&
Alerting Tests

UREQ_3, FREQ_5,
TREQ_6

AREA 04

User Interface Tests

UREQ_2, FREQ_4,
TREQ_7

HARDWARE & SIGNAL ACQUISITION TESTS

Test ID	Procedure	Pass Criteria
T1.1	Acquire IQ samples from the antenna for 10 seconds at rate 10e6, verify the output buffer is non-zero	Zero dropped samples or buffer overflows during continuous capture
T1.2	Simultaneously acquire position and raw IQ data	Successful location data stream alongside IQ snapshots in <60s
T1.3	Start the system and let it run for 1 hour without human intervention	100% uptime over the time window
T1.4	Power on the device and wait for it to reach "ACTIVE" status on the UI	System ready for monitoring in < 60 seconds from power-on

AI MODEL & CLASSIFICATION TESTS

Test ID	Procedure	Pass Criteria
T2.1	Inject 500 clean GNSS signal snapshots and record the AI model output class	≥95% of snapshots classified as clean
T2.2	Generate jamming with HackRF at specific JSR levels (LOW, MID, HIGH) for all Jamming types (LWF, LN, TRI, TRIW, TICK), run segments through the pipeline, record outputs	Per-class accuracy ≥90%
T2.3	Measure the full pipeline time over 500 snapshot	Mean task completion ≤ 100 ms, to ensure real-time operation

COMMUNICATION & ALERTING TESTS

Test ID	Procedure	Pass Criteria
T3.1	Trigger a simulated jamming event on the Processing Unit and measure time to UI notification	End-to-end alert delivery with 10/10 alerts delivered and mean latency ≤ 5 s
T3.2	Forcibly disconnect the device (power cut) to verify that the HiveMQ broker publishes Last Will Testament message to the Control Center	Control Center UI receives the LWT message and changes the truck status to "OFFLINE" in the Active Vehicles Table
T3.3	Simulate a brief network interruption during a jamming event and verify that the alert is re-delivered once connectivity is restored (QoS level 1 guarantee)	Alert is queued and delivered to Control Center after reconnection without user action in 10/10 tests.
T3.4	Connect two simulated devices with different vehicle plate IDs and trigger jamming only on 1 device and verify that the Control Center receives the alert only on the correct topic	0 messages on the wrong topic in 10 test events

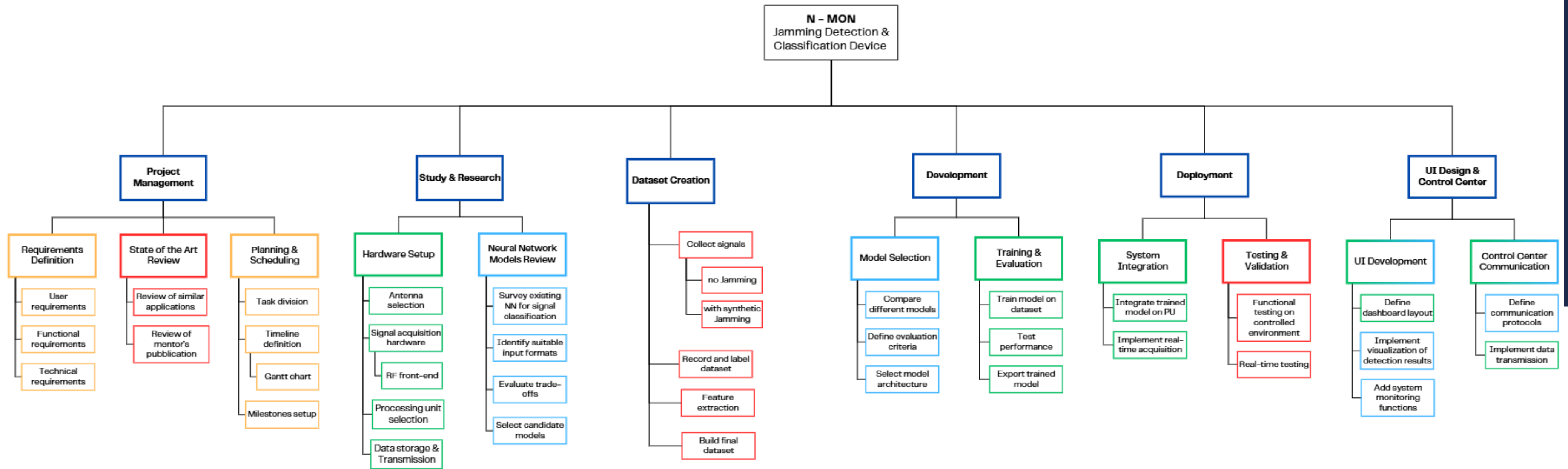
USER INTERFACE TESTS

Test ID	Procedure	Pass Criteria
T4.1	Open Control Center UI and the Driver mobile UI on separate devices and measure load time from first network request to fully rendered page	Load time $\leq 5s$ on both desktop and mobile browser in 10/10 tests
T4.2	With the UI open, trigger a jamming event and measure the time from MQTT message publication to the moment the jamming counter increments and the pop-up alert appears in the browser	UI update latency $\leq 5s$ in 10/10 tests
T4.3	Trigger 5 distinct jamming events with different jamming types and verify that all 5 appear in the jamming event table with correct Plate, Jamming Type, Time, and Position fields.	All 5 rows populated correctly with no missing or incorrect fields
T4.4	Populate Alert Table and apply filters for all possible parameters on the dashboard	Only events matching the filter criteria are displayed both in the Alert Table and Map in 10 filter test cases

Link accessible with Polito credentials :

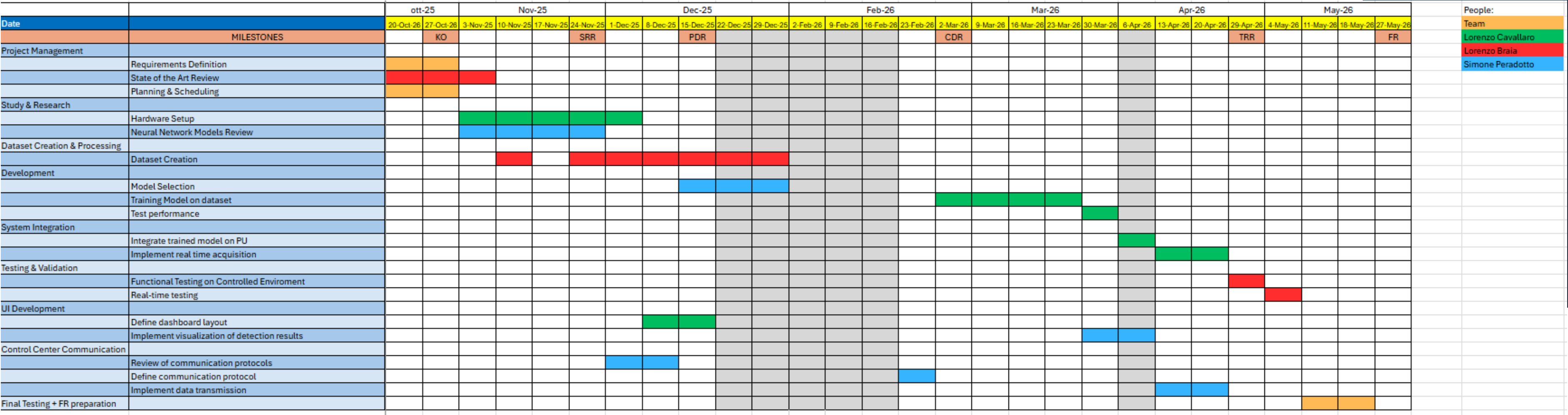
https://politoit-my.sharepoint.com/:x:/g/personal/s343420_studenti_polito_it/EVHoF8byxhZOuSA_gHzWz6ABPUtiU81AzwQXv3SZYcAgyA?e=vecgTL

WBS



<https://www.canva.com/design/DAG14EVzFeY/AZdbRfanr0YUQmWhZWHVmg/edit>

GANTT CHART



Link accessible with Polito credentials :

https://politoit-my.sharepoint.com/:x:/g/personal/s343420_studentsi_polito_it/EVHoF8byxhZOuSA_gHzWz6ABPUtiU81AzwQXv3SZYcAgyA?e=vecgTL



THANK YOU!

For your attention